ALZ-PATHWAY-TWIN

Validation Report. Three benchmark tasks, six pre-declared predictions and the trials the model was fitted to: what worked and what did not.

0.901
ROC AUC, five-year dementia forecast, logistic regression

0.659
PEHE of the best effect estimator, against 0.673 for a constant effect

0 of 5
pre-declared predictions matched; one withdrawn

0.04
power to separate pre-symptomatic shares in a symptomatic trial

Key results

task 1 AUC (LR) **0.901**	task 1 AUC (XGBoost) **0.895**	task 1 Brier (LR) **0.0123**	p-tau217 only AUC **0.829**
task 2 PEHE ridge **0.659**	task 2 PEHE XGBoost **0.837**	true mean effect **-0.150 CDR-SB**	trial pool **6,335 people**

A synthetic dataset is only as useful as the questions it can honestly answer. This report lists the questions it answers and the ones it does not.

All data are synthetic and describe no real person. Effect shares in the scenario grid are hypotheses, not measured quantities, and nothing here is medical advice or a substitute for clinical evidence. The tables report where the model does not match published numbers, and the passport lists them.

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01 Summary

The report has three benchmark tasks and a table of predictions. For each task: what was asked, which model was used, the metrics inside the dataset, and the comparison with a published metric on real data, where one exists. Where no published analogue exists, the report says so.

WHAT WORKS

Standard prediction pipelines run on the data and give plausible scores. A five-year dementia forecast from the entry visit reaches AUC 0.90, and plasma p-tau217 alone reaches 0.83, close to the published 0.83 for a different population. The paired design allows scoring of individual-effect estimators, which real data cannot.

WHAT DOES NOT WORK

None of the five pre-declared predictions matched within its tolerance: donanemab slowing came out at 0.27 against 0.36 published, subtype survival came out almost equal across subtypes, amyloid positivity at age 90 at 0.30 against 0.44, and MCI-to-dementia conversion at 0.19 per year against 0.05 to 0.10. One further prediction (acute amyloid rise after sleep loss) was withdrawn because the model has no sleep state. A symptomatic 18-month trial cannot separate pre-symptomatic effect shares at all.

02 Task 1: five-year dementia forecast

Task. From the entry visit of the untreated scenario S00, forecast dementia (CDR-SB 4.5) within five years. People with dementia at entry are excluded; people without dementia who died or ended follow-up before year five are dropped. Result: 8,577 people, 183 events (2.1 %). Features: age, sex, APOE e4 alleles, MMSE, CDR-SB, log p-tau217.

Models. Logistic regression with standardized inputs; gradient-boosted trees (300 trees, depth 3, learning rate 0.05). Five-fold stratified cross-validation, seed 20261108.

Model and features	ROC AUC	Brier
Logistic regression, all features	0.901	0.0123
XGBoost, all features	0.895	0.0125
Logistic regression, p-tau217 + MMSE + e4	0.882	0.0163
Logistic regression, p-tau217 only	0.829	0.0203

Event rate 2.1 %, so a Brier score of 0.0123 should be read against the no-skill value of about 0.0209.

Comparison with published results. A published study of people with subjective cognitive decline and mild cognitive impairment reports that "plasma P-tau217 predicted AD accurately (area under the curve (AUC) = 0.83)" within four years, and 0.91 when combined with memory, executive function and APOE (Nature Medicine, 2021, DOI 10.1038/s41591-021-01348-z). A Swedish community cohort reports AUCs from 70.9 % to 82.6 % for 10-year dementia (Nature Medicine, 2025, DOI 10.1038/s41591-025-03605-x). The synthetic values (0.83 and 0.90) are in the same range, but the comparison is not like for like: populations, horizons and outcomes differ, and the synthetic cohort has no dementia other than the Alzheimer type, which makes forecasting easier.

03 Task 2: individual effect of early treatment

Task. Half of the cohort is assigned at random to S15 (screening plus lecanemab), half to S00, as in a randomized trial. From entry-visit features, a T-learner estimates the difference in CDR-SB at year 10. The true individual effect comes from the paired trajectory of the same person. Result: 7,617 people alive at year 10 in both worlds; true mean effect -0.150 CDR-SB points; spread of individual effects 0.673.

Models. T-learner with ridge regression (the continuous-outcome counterpart of the logistic model) and with gradient-boosted trees; five-fold cross-fitting.

Estimator	Mean-effect bias	PEHE	Rank correlation
Ridge T-learner	0.0056	0.659	0.31
XGBoost T-learner	0.0008	0.837	0.12
Constant effect (reference)	0	0.673	-
Zero effect (reference)	0.1495	0.690	-

PEHE below 0.673 means the estimator found some individual structure beyond the mean.

Comparison. No published analogue exists for a synthetic counterfactual task. Reading: both estimators recover the mean effect almost exactly, the ridge estimator beats the constant-effect reference only slightly (0.659 against 0.673), and the tree model is worse than the constant (0.837): a 10-year effect of this size, hidden in noisy observed CDR-SB, is hard to localise from entry features with 3,800 people per arm.

04 Task 3: can a trial tell the hypotheses apart?

Task. 1,000 virtual 18-month trials of 900 people per arm are drawn from a pool of 6,335 eligible virtual participants (amyloid positive, CDR-SB 3.0 to 3.4 at entry). For each pair of neighbouring grid settings, power is the share of trials in which the CDR-SB change differs significantly at the 0.05 level. This is a trial-power calculation; no machine learning is involved.

Pair of neighbouring settings	Difference in mean change	Power
Clearance share 0 vs 0.5 of the amyloid share	0.042	0.092
Clearance share 0.5 vs 1.0	0.042	0.116
Inflammation share 0 vs mid	0.293	0.999
Inflammation share mid vs high	0.303	1.000
Amyloid share 0.5 vs 1.0 of the fitted value	0.228	0.997
Amyloid share 1.0 vs 1.5	0.234	0.998
Pre-symptomatic amyloid share 0.3 vs 0.9 (symptomatic share fixed)	0.000	0.042

Power of 0.05 is the false-positive rate, so a value near it means the trial cannot tell the settings apart.

FOR TRIAL PLANNERS

An 18-month trial of 900 per arm separates the inflammation and amyloid settings of the grid almost surely, but cannot separate the clearance settings (power 0.09 and 0.12), because the model's clearance mechanism changes the CDR-SB change by only 0.04 points. It also cannot separate pre-symptomatic shares at all: the pre-symptomatic share acts only before CDR-SB 0.5, and participants of a symptomatic trial are past that point, so arms that differ only in that share have identical expected change (difference 0.000, power 0.042).

05 Pre-declared predictions

ID	Prediction	Published	Predicted	Result	Likely reason
P1	Donanemab slowing of CDR-SB, low/medium tau	0.362	0.269	no match	Donanemab clears amyloid more deeply than lecanemab, but the model reuses the lecanemab mechanism; treated change 1.42 against 1.20 published
P2	Survival from dementia onset by subtype (3 shortest, 1 longest)	5.6 y / 8.9 y	3.2-3.9 y	no match	Subtypes differ in the model only by tau rate and genotype mix; survival after dementia onset has no subtype term
P3	Amyloid rise after one night without sleep	~5 %	withdrawn	withdrawn	The model has no sleep state; expressing the prediction would have needed a new free parameter
P4	Amyloid positivity at age 90, unimpaired	0.44	0.302	no match	Onset ages were fitted at age 50 and to the carrier ratio; the late-life plateau of positivity is not reproduced
P5	Order of 12-month MCI-to-dementia conversion by subtype	order, not significant	0.184-0.195	no match	Subtypes barely differ in conversion (0.184 to 0.195); the model has no subtype term for the speed of conversion
P6	Annual MCI-to-dementia conversion, natural course	0.049 / 0.096	0.192	compared, no threshold	The model has no non-Alzheimer dementias and no non-amyloid MCI, and the populations differ

The predictions were written down and sealed before the full run; results are recorded as they came out. The reasons are the builder's analysis, not tested hypotheses.

06 Honest limitations of this validation

- **The benchmark is internal.** Scores on synthetic data say the pipeline works on the data; they do not say a model would work on real patients.
- **Comparisons with published metrics are not like for like** (populations, horizons, outcomes differ) and are given as context only.
- **Task 1 has 183 events,** so AUC differences of 0.01 are within noise; the difference between 0.90 and 0.83 is not.
- **Task 2 uses one effect shape** (S15 against S00 at year 10); other scenarios and horizons may behave differently.
- **Task 3 powers are conditional** on the model's own noise (CDR-SB test-retest SD 0.5) and on the pool of virtual participants.
- **Task 3 cannot be recomputed from the tables alone.** It re-simulates the pool of participants under settings that are not in the scenario grid, so its numbers are reported as computed.
- **The four theories are not tested against each other** by this report; it shows only how separable their grid settings are.

PART II · TECHNICAL SPECIFICATION

07 Replication of the fitted trials

Replication	Endpoint	Published	Simulated	Dev. %	n
Lecanemab trial, amyloid PET	Treatment-minus-placebo change in amyloid PET at 18 months, centiloid	-59.1	-59.1	0.0	4258
Lecanemab trial, CDR-SB placebo arm	CDR-SB change at 18 months, placebo-like arm	1.66	1.6633	0.2	4258
Lecanemab trial, CDR-SB slowing	Slowing of CDR-SB change at 18 months (0.45 / 1.66)	0.2711	0.2711	0.002	4258

Lecanemab: Lecanemab in Early Alzheimer's Disease, N Engl J Med 2023, DOI 10.1056/NEJMoa2212948. Soluble-TNF trial: XPro1595 in Early Alzheimer's Disease with Inflammation (MINDFuL), medRxiv 2025, DOI 10.1101/2025.09.24.25336496. These agree by construction: the model parameters were fitted to them.

Replication	Endpoint	Published	Simulated	Dev. %	n
XPro1595 MINDFuL trial, amyloid-positive early AD with inflammation	Standardized effect on cognitive change at 24 weeks	0.27	0.27	0.0	4015

08 Calibration of the five-year forecast

Decile	Mean predicted	Observed share	n
00	0.0008	0.0000	858
01	0.0016	0.0012	858
02	0.0023	0.0023	857
03	0.0032	0.0035	858
04	0.0043	0.0023	858
05	0.0058	0.0093	857
06	0.0081	0.0093	858
07	0.0117	0.0128	857
08	0.0200	0.0221	858
09	0.1551	0.1503	858

Logistic regression, all features. The top decile predicts 0.155 and observes 0.150; the eight lower deciles are noisy because they hold between 0 and 19 events each.

09 Reproducing the benchmark

Tasks 1 and 2 are reproduced from the data tables with the benchmark script; they need scikit-learn and xgboost. The script writes the metrics of both tasks to a Parquet table.

```
python3 benchmark_script.py --data <data_dir> --out benchmark_results.parquet
```

Task 3 cannot be recomputed from the data tables: it re-simulates a pool of 60,000 virtual people under settings that are not in the scenario grid, so its results are reported as computed (seed 31337 for the pool, 1,000 trials of 900 people per arm).

10 Defects found and fixes applied

Defect found	Fix applied
A first calibration of the amyloid share used a noisier 16,000-person run.	Re-run on 48,000 people; the reported deviations come from that run.
A draft of the prediction register named authors without a retrieved record.	Replaced by titles, journals, years and DOIs taken from retrieved records; the register was re-sealed before the run.
Task 3 settings varied the amyloid share at both stages together, which could suggest the pre-symptomatic share is testable in a trial.	Added the explicit test of a pre-symptomatic-only difference: difference 0.000, power 0.042.
A pre-declared prediction (sleep) referred to a mechanism the model does not have.	Withdrawn before the run and listed as withdrawn, not as passed or failed.

11 Assumptions log

Assumption	Basis
Dementia is CDR-SB of 4.5 or more	Published CDR-SB staging ranges
Observed CDR-SB noise SD 0.5 applies to every visit	Published test-retest reliability, converted
A trial arm is drawn from a pool of CDR-SB 3.0-3.4, amyloid-positive people	Matches the published lecanemab trial baseline
Welch t-test at 0.05 stands for the trial analysis	Simplification; real trials use mixed models

12 Which questions the data can answer

Question	Answer	Basis
Does my effect estimator recover a known individual effect?	Yes	Paired trajectories (task 2)
Can an 18-month trial separate amyloid or inflammation settings?	Yes	Power 0.997 and above (task 3)
Can it separate clearance settings or pre-symptomatic shares?	No	Power 0.09 to 0.12 and 0.04 (task 3)
Which mechanism theory is true?	Not answered	All grid worlds share one natural course
Will a model trained here work on real patients?	Not answered	Internal benchmark only

13 Reading the scores

ROC AUC measures ranking: the chance that a person who develops dementia within five years gets a higher forecast than one who does not. The Brier score measures calibration and sharpness together; with a 2.1 % event rate a model that always predicts the base rate scores about 0.0209, so 0.0123 is a real improvement. PEHE is the typical size of the error in the estimated individual effect; the reference of 0.673 is the error of a model that knows the true mean effect but nothing individual.

14 Terms used

Term	Meaning
CDR-SB	Clinical Dementia Rating sum of boxes, 0 to 18; higher is worse. 0.5 marks the first symptoms (mild cognitive impairment stage here) and 4.5 the start of dementia.
MMSE	Mini-Mental State Examination, 0 to 30; higher is better.
p-tau217	Plasma concentration of tau phosphorylated at threonine 217, in pg/mL; two cut points define negative, intermediate and positive zones.
centiloid (CL)	Standard scale of amyloid PET signal; 0 is the young-control level and 100 the typical Alzheimer level. Positivity is set at 20 CL.
ARIA-E	Amyloid-related imaging abnormality with oedema, a side effect of anti-amyloid antibodies.
APOE e4	Gene variant that raises the risk of Alzheimer's disease; carried as 0, 1 or 2 alleles.
PEHE	Root mean squared error of the individual treatment effect estimate.